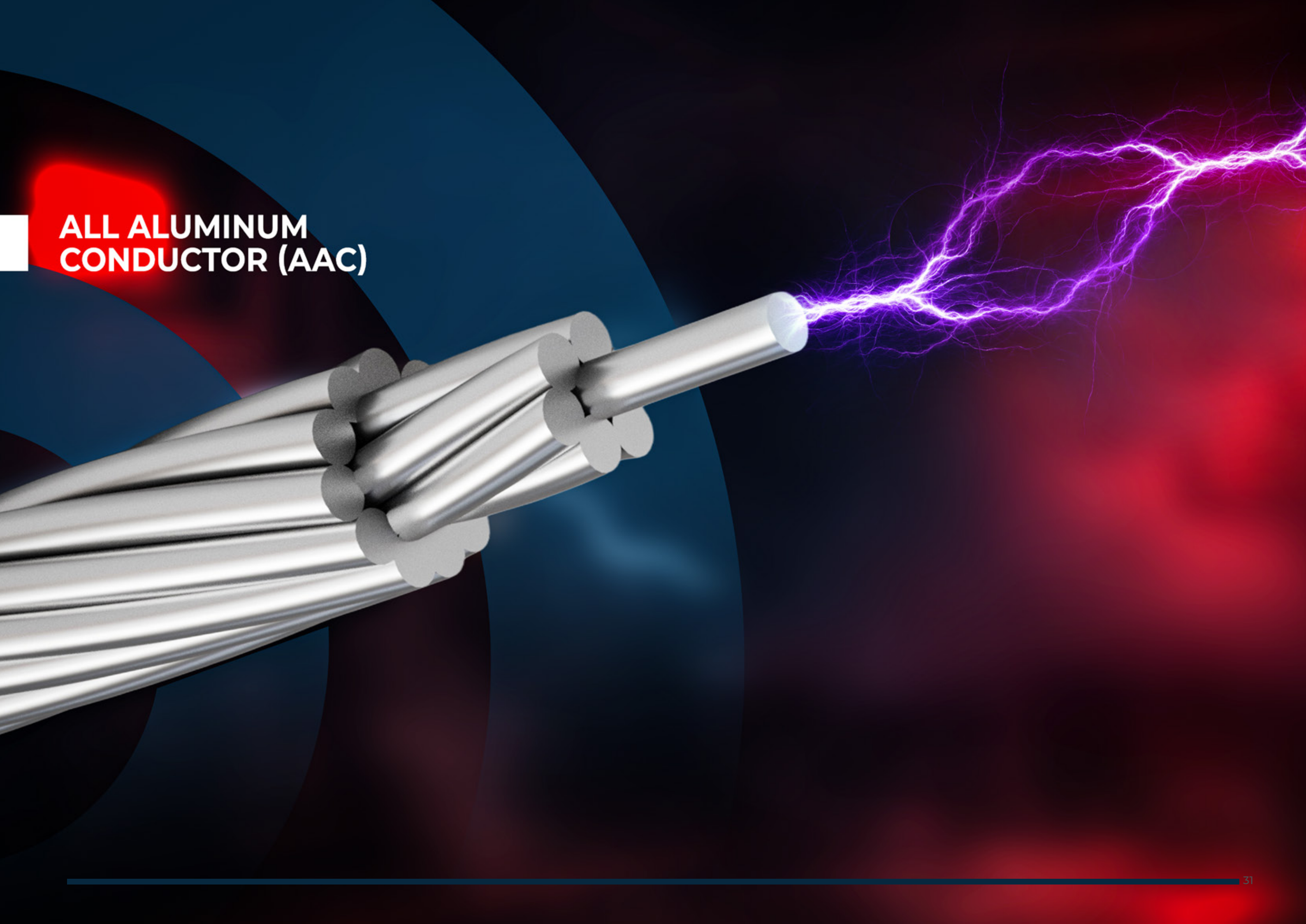


A 3D rendering of a bundle of aluminum conductors (AAC) against a dark blue and red background. The bundle consists of several cylindrical rods. One rod is extended further than the others, and a bright purple lightning bolt strikes its tip, creating a dynamic visual effect. The background features large, curved shapes in dark blue and red, with a glowing red area on the left.

ALL ALUMINUM CONDUCTOR (AAC)

All Aluminum Conductors
(AAC)



Description

Hard drawn Aluminum wires , stranded in successive layers, in opposite direction to form the Aluminum stranded AAC conductor.

Application

All Aluminum Conductors (AAC) can be used as a bare overhead conductor for distribution lines. Because of its low strength-to-weight ratio, (AAC) has limited use in transmission lines. AAC is extensively used in urban areas where spans are usually short but high conductivity is required.

Technical Data

Relevant Standard	BS EN 50182, BS 215-1, DIN 48201-5, IEC 61089, ASTM B 231
Conductor	Plain bare hard drawn Aluminum - H14
Minimum bending Radius	25 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request

Product Code	Cross Sectional Area mm ²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm ²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
A - BS EN 50182 (GERMANY) & DIN 48201-5							
PO-BT11-00-01-U0	16	7 x 1.70	1.7986	5.10	3.02	43.4	110
PO-BT12-00-01-U0	25	7 x 2.10	1.1787	6.30	4.36	66.3	145
PO-BT13-00-01-U0	35	7 x 2.50	0.8317	7.50	6.01	93.9	180
PO-BT14-00-01-U0	50	7 x 3.00	0.5776	9.00	8.41	135.2	225
PO-BT14-00-01-U0	50	19 x 1.80	0.5944	9.00	8.94	132.9	225
PO-BT15-00-01-U0	70	19 x 2.10	0.4367	10.50	11.85	180.9	270
PO-BT16-00-01-U0	95	19 x 2.50	0.3081	12.50	16.32	256.3	340
PO-BT17-00-01-U0	120	19 x 2.80	0.2456	14.00	19.89	321.5	390
PO-BT18-00-01-U0	150	37 x 2.25	0.1960	15.80	26.48	405.7	455
PO-BT19-00-01-U0	185	37 x 2.50	0.1588	17.50	31.78	500.9	520
PO-BT20-00-01-U0	240	61 x 2.25	0.1193	20.30	43.66	671.1	625
PO-BT21-00-01-U0	300	61 x 2.50	0.0966	22.50	52.40	828.5	710
PO-BT22-00-01-U0	400	61 x 2.89	0.0723	26.00	68.02	1107.1	855
PO-BT23-00-01-U0	500	61 x 3.23	0.0579	29.10	82.47	1382.9	990

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

All Aluminum Conductors
(AAC)



Product Code	Cross Sectional Area mm ²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm ²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
B - BS EN 50182 (United Kingdom)							
MIDGE	23.3	7 x 2.06	1.2249	6.18	4.20	63.8	139
GNAT	26.9	7 x 2.21	1.0643	6.63	4.83	73.4	153
MOSQUITO	36.9	7 x 2.59	0.7749	7.77	6.27	100.8	187
LADYBIRD	42.8	7 x 2.79	0.6678	8.37	7.28	117.0	205
ANT	52.8	7 x 3.1	0.5409	9.30	8.72	144.4	234
FLY	63.6	7 x 3.4	0.4497	10.20	10.49	173.7	264
BLUEBOTTLE	73.6	7 x 3.66	0.3880	11.00	11.78	201.3	291
EARWING	78.6	7 x 3.78	0.3638	11.30	12.57	214.7	302
GRASSHOPPER	84.1	7 x 3.91	0.3400	11.70	13.45	229.7	315
CLEGG	95.6	7 x 4.17	0.2989	12.50	15.30	261.3	343
WASP	106	7 x 4.39	0.2697	13.20	16.95	289.6	366
BEETLE	106.4	19 x 2.67	0.2701	13.40	18.08	292.4	368
BEE	132	7 x 4.90	0.2165	14.70	21.12	360.8	421
HORNET	157.6	19 x 3.25	0.1823	16.30	26.01	433.2	473
CATERPILLER	185.9	19 x 3.53	0.1546	17.70	29.75	511.1	526
CHAFER	213.2	19 x 3.78	0.1348	18.90	34.12	586.0	574
SPIDER	237.6	19 x 3.99	0.1210	20.00	38.01	652.9	615
COCKROACH	265.7	19 x 4.22	0.1081	21.10	42.52	730.0	660
BUTTERFLY	322.7	19 x 4.65	0.0891	23.30	51.63	886.8	748
DRONE	372.4	37 x 3.58	0.0774	25.10	59.59	1027.1	821
CENTIPEDE	415.2	37 x 3.78	0.0695	26.50	66.43	1145.1	879
MAYBUG	486.1	37 x 4.09	0.0593	28.60	77.78	1340.6	972
SCORPION	529.8	37 x 4.27	0.0544	29.90	84.77	1461.2	1026
CICADA	628.3	37 x 4.65	0.0459	32.60	100.54	1732.9	1143

The above data is approximate and subject to manufacturing tolerances



Product Code	Cross Sectional Area mm ²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm ²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
C - IEC 61089							
A1 - 16	16	7 x 1.71	1.7896	5.12	3.04	43.8	146
A1 - 25	25	7 x 2.13	1.1453	6.40	4.50	68.4	194
A1 - 40	40	7 x 2.70	0.7158	8.09	6.80	109.4	262
A1 - 63	63	7 x 3.39	0.4545	10.20	10.39	172.3	351
A1 - 100	100	19 x 2.59	0.2877	12.90	17.00	274.8	472
A1 - 125	125	19 x 2.89	0.2302	14.50	21.25	343.6	545
A1 - 160	160	19 x 3.27	0.1798	16.40	26.40	439.8	638
A1 - 200	200	19 x 3.66	0.1439	18.30	32.00	549.7	737
A1 - 250	250	19 x 4.09	0.1151	20.50	40.00	687.1	852
A1 - 315	315	37 x 3.29	0.0916	23.00	51.97	867.9	987
A1 - 400	400	37 x 3.71	0.0721	26.00	64.00	1102.0	1152
A1 - 450	450	37 x 3.94	0.0641	27.50	72.00	1239.8	1240
A1 - 500	500	37 x 4.15	0.0577	29.00	80.00	1377.6	1327
A1 - 560	560	37 x 4.39	0.0515	30.70	89.60	1542.9	1425
A1 - 630	630	61 x 3.63	0.0458	32.60	100.80	1738.3	1535
A1 - 710	710	61 x 3.85	0.0407	34.60	113.60	1959.1	1653
A1 - 800	800	61 x 4.09	0.0361	36.80	128.00	2207.4	1783
A1 - 900	900	61 x 4.33	0.0321	39.00	144.00	2483.3	1914
A1 - 1000	1000	61 x 4.57	0.0289	41.00	160.00	2759.2	2039

Note:-

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

All Aluminum Conductors
(AAC)



Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
D - BS 215-1							
MIDGE	22	7 x 2.06	1.2270	6.18	3.99	64	139
ANT	50	7 x 3.10	0.5419	9.30	8.28	145	234
FLY	60	7 x 3.40	0.4505	10.20	9.90	174	264
WASP	100	7 x 4.39	0.2702	13.17	16.00	290	366
HORNET	150	19 x 3.25	0.1825	16.25	25.70	434	472
CHAFER	200	19 x 3.78	0.1349	18.90	32.40	587	573
COCKROACH	250	19 x 4.22	0.1083	21.10	40.40	731	660
BUTTERFLY	300	19 x 4.65	0.0892	23.25	48.75	888	748
CENTIPEDE	400	37 x 3.78	0.0694	26.46	63.10	1145	879
Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
E - ASTM B 231							
PEACHBELL	13.3	7 x 1.56	2.1402	4.68	2.53	36.6	98
ROSE	21.1	7 x 1.96	1.3558	5.88	3.91	58.2	130
IRIS	33.6	7 x 2.47	0.8537	7.41	5.99	92.6	175
PANSY	42.4	7 x 2.78	0.6740	8.34	7.30	116.6	204
POPPY	53.5	7 x 3.12	0.5351	9.36	8.84	147.2	236
ASTER	67.4	7 x 3.50	0.4252	10.50	11.10	185.7	273
PHLOX	85	7 x 3.93	0.3373	11.79	13.50	233.9	317
OXLIP	107.2	7 x 4.42	0.2666	13.26	17.00	295.2	369
VALERIAN	126.7	19 x 2.91	0.2277	14.55	20.70	348.6	410
SNEEZEWORT	126.7	7 x 4.80	0.2261	14.40	20.10	348.8	410
LAUREL	135.2	19 x 3.01	0.2129	15.05	22.10	372.2	428
PEONY	152	19 x 3.19	0.1895	15.95	24.30	418.3	461
TULIP	170.5	19 x 3.38	0.1688	16.90	27.30	469.5	497

Note:-

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

DAFFODIL	177.3	19 x 3.45	0.1620	17.25	28.40	487.9	510
CANNA	201.4	19 x 3.67	0.1432	18.35	31.60	554.9	552
GOLDENTUFT	228	19 x 3.91	0.1262	19.55	35.00	627.6	599
SYRINGA	241.7	37 x 2.88	0.1197	20.16	38.60	664.8	620
COSMOS	241.7	19 x 4.02	0.1194	20.10	37.00	664.8	620
HYACINTH	253.3	37 x 2.95	0.1141	20.65	40.50	696.8	640
ZINNIA	253.3	19 x 4.12	0.1136	20.60	38.90	697.1	640
MISTLETOE	282	37 x 3.12	0.1020	21.84	44.30	775.7	687
DAHLIA	282	19 x 4.35	0.1019	21.75	43.30	775.8	687
MEADOWSWEET	304	37 x 3.23	0.0952	22.61	47.50	836.3	718
ORCHID	322.3	37 x 3.33	0.0893	23.31	50.40	886.9	748
HEUCHERA	329.4	37 x 3.37	0.0874	23.59	51.70	907.4	759
FLAG	354.7	61 x 2.72	0.0815	24.48	57.10	975.8	793
VERBENA	354.7	37 x 3.49	0.0815	24.43	55.40	975.7	793
NASTURTIUM	362.6	61 x 2.75	0.0798	24.75	58.40	998.5	805
VIOLET	362.6	37 x 3.53	0.0797	24.71	56.70	998.5	805
CATTAIL	380	61 x 2.82	0.0759	25.38	60.30	1046.0	831
PETUNIA	380	37 x 3.62	0.0758	25.34	58.60	1046.0	831
LILAC	402.8	61 x 2.90	0.0717	26.10	63.80	1110.0	860
ARBUTUS	402.8	37 x 3.72	0.0718	26.04	61.80	1109.0	860
SNAPDRAGON	456	61 x 3.09	0.0632	27.81	70.80	1256.0	932
COCKSCOMB	456	37 x 3.96	0.0633	27.72	68.40	1256.0	932
GOLDENNROD	483.4	61 x 3.18	0.0597	28.62	75.00	1331.0	968
MAGNOLIA	483.4	37 x 4.08	0.0597	28.56	72.60	1331.0	968
CAMELLIA	506.7	61 x 3.25	0.0571	29.25	78.30	1394.0	996
HAWKWEED	506.7	37 x 4.18	0.0569	29.26	76.20	1395.0	996
LARKSPUR	523.7	61 x 3.31	0.0551	29.79	81.30	1442.0	1019
BLUEBELL	523.7	37 x 4.25	0.0550	29.75	78.80	1441.0	1019
MARIGOLD	564	61 x 3.43	0.0513	30.87	87.30	1553.0	1066
HAWTHORM	604.2	61 x 3.55	0.0479	31.95	93.50	1662.0	1113
NARCISSUS	644.5	61 x 3.67	0.0448	33.03	98.10	1774.0	1161
COLUMBINE	694.8	61 x 3.78	0.0422	34.02	104.00	1884.0	1205
CARNATION	725.1	61 x 3.89	0.0399	35.01	108.00	1997.0	1247
GLADIOLUS	765.4	61 x 4.00	0.0377	36.00	114.00	2108.0	1292
COREOPSIS	805.7	61 x 4.10	0.0359	36.90	120.00	2216.0	1331
JASSAMINE	886.7	61 x 4.30	0.0327	38.70	132.00	2442.0	1410

The above data is approximate and subject to manufacturing tolerances